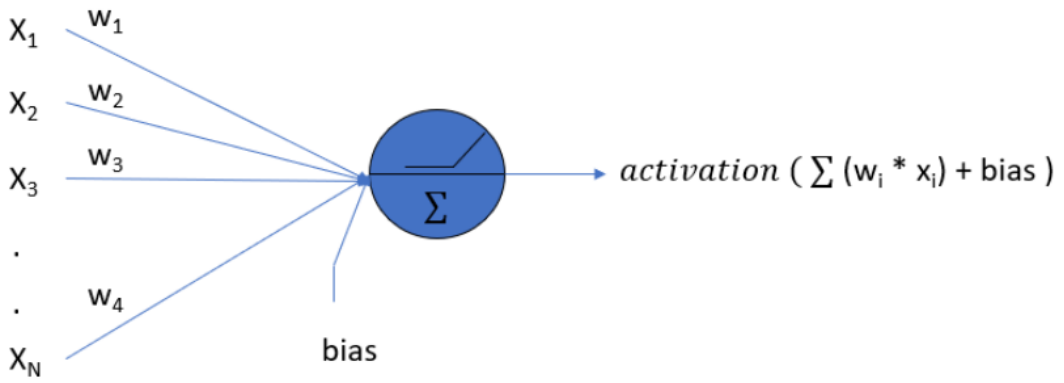
Neural Networks: Multilayer perceptions, Back- propagation algorithm, Training strategies, Activation Functions, Gradient Descent for Machine Learning, Radialbasis functions, Hopfield network, Recurrent Neural Networks.

Deep Learning: Introduction to deep learning, Convolution Neural Networks (CNN), CNN Architecture, pre-trained CNN (LeNet, AlexNet).

Neural networks are used to mimic the basic functioning of the human brain and are inspired by how the human brain interprets information. It is used to solve various real-time tasks because of its ability to perform computations quickly and its fast responses.



A single neuron shown with X_i inputs with their respective weights W_i and a bias term and applied activation function

Types of Neural Networks

(i) ANN– It is also known as an artificial neural network. It is a feed-forward neural network because the inputs are sent in the forward direction. It can also contain hidden layers which can make the model even denser. They have a fixed length as specified by the programmer. It is used for Textual Data or Tabular Data. A widely used real-life application is Facial Recognition. It is comparatively less powerful than CNN and RNN.

(ii) CNN– It is also known as Convolutional Neural Networks. It is mainly used for Image Data. It is used for Computer Vision. Some of the real-life applications are object detection in

autonomous vehicles. It contains a combination of convolutional layers and neurons. It is more powerful than both ANN and RNN.

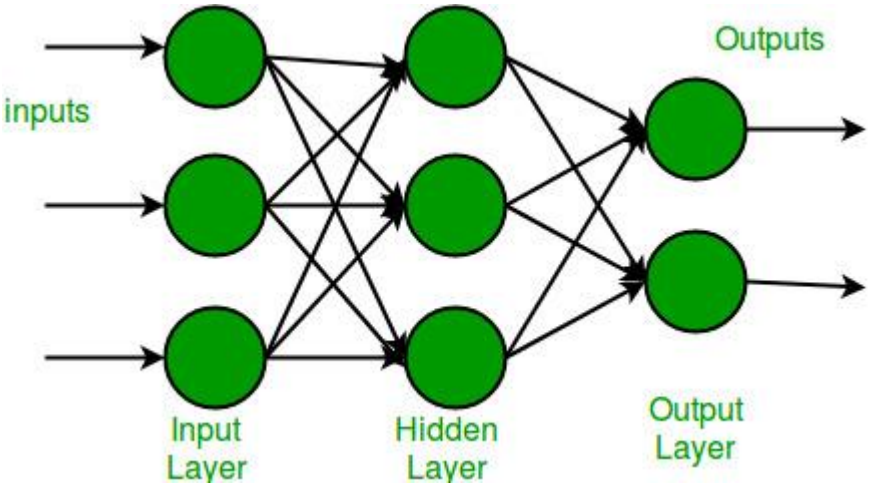
(iii) RNN–

It is also known as Recurrent Neural Networks. It is used to process and interpret time series data. In this type of model, the output from a processing node is fed back into nodes in the same or previous layers. The most known types of RNN are **LSTM** (Long Short Term. Memory) Networks.

Multi-layer Perceptron

Multi-layer perception is also known as MLP. It is fully connected dense layers, which transform any input dimension to the desired dimension. A multi-layer perception is a neural network that has multiple layers. To create a neural network we combine neurons together so that the outputs of some neurons are inputs of other neurons.

A multi-layer perceptron has one input layer and for each input, there is one neuron(or node), it has one output layer with a single node for each output and it can have any number of hidden layers and each hidden layer can have any number of nodes. A schematic diagram of a Multi-Layer Perceptron (MLP) is depicted below.



Every node in the multi-layer perception uses a sigmoid activation function. The sigmoid activation function takes real values as input and converts them to numbers between 0 and 1 using the sigmoid formula.

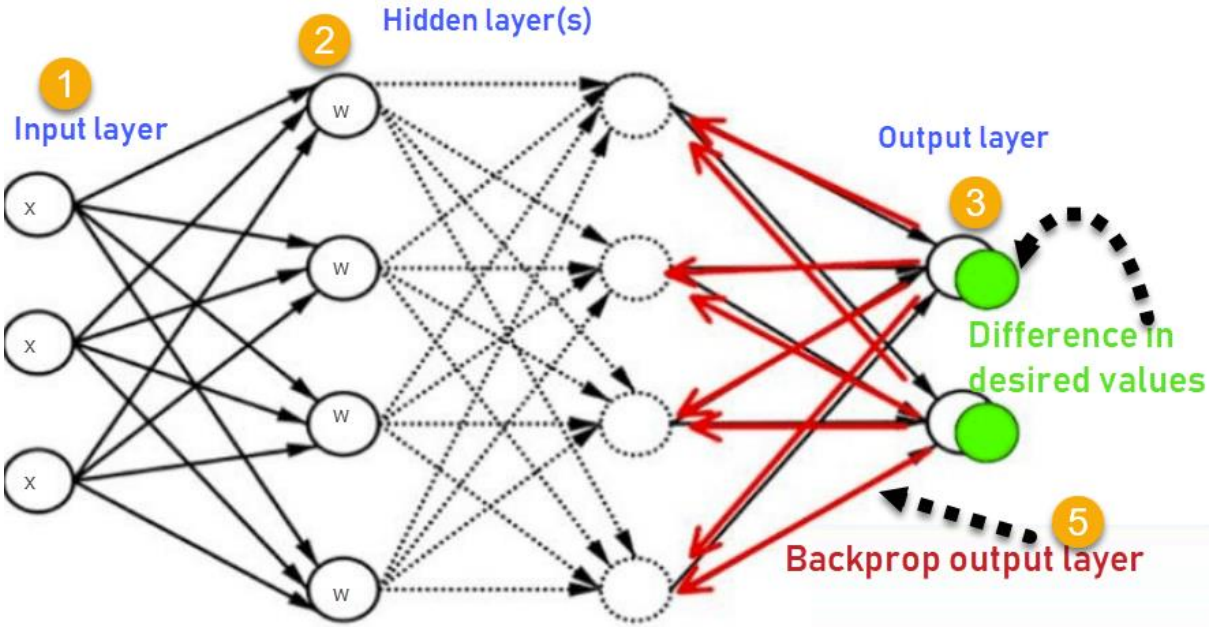
Backpropagation

is the essence of neural network training. It is the method of fine-tuning the weights of a neural network based on the error rate obtained in the previous epoch (i.e., iteration). Proper tuning of the weights allows you to reduce error rates and make the model reliable by increasing its generalization.
Backpropagation in neural network is a short form for "backward propagation of errors." It is a standard method of training artificial neural networks. This method helps calculate the gradient of a loss function with respect to all the weights in the network.

Backpropagation Algorithm Works

The Back propagation algorithm in neural network computes the gradient of the loss function for a single weight by the chain rule. It efficiently computes one layer at a time, unlike a native direct computation. It computes the gradient, but it does not define how the gradient is used. It generalizes the computation in the delta rule.

Consider the following Back propagation neural network example diagram to understand:



1. Inputs X, arrive through the preconnected path
2. Input is modeled using real weights W. The weights are usually randomly selected.
3. Calculate the output for every neuron from the input layer, to the hidden layers, to the output layer.
4. Calculate the error in the outputs

$Error_B$= Actual Output – Desired Output

5. Travel back from the output layer to the hidden layer to adjust the weights such that the error is decreased.

Keep repeating the process until the desired output is achieved

Types of Backpropagation Networks

Two Types of Backpropagation Networks are:

- Static Back-propagation
- Recurrent Backpropagation

Static back-propagation:

It is one kind of backpropagation network which produces a mapping of a static input for static output. It is useful to solve static classification issues like optical character recognition.

Recurrent Backpropagation:

Recurrent Back propagation in data mining is fed forward until a fixed value is achieved. After that, the error is computed and propagated backward.

The main difference between both of these methods is: that the mapping is rapid in static back-propagation while it is nonstatic in recurrent backpropagation.

Training strategy

The procedure used to carry out the learning process is called the training (or learning) strategy. The training strategy is applied to the neural network to obtain the minimum loss possible. This is done by searching for parameters that fit the neural network to the data set.

A general strategy consists of two different concepts:

- **Loss index**
- **Optimization algorithm**

Loss index

The loss index plays a vital role in the use of neural networks. It defines the task the neural network is required to do and provides a measure of the quality of the representation required to learn. The choice of a suitable loss index depends on the application.

When setting a loss index, two different terms must be chosen: an error term and a regularization term.
0

$$loss\_index = error\_term + regularization\_term$$

Error term

The error is the most important term in the loss expression. It measures how the neural network fits the data set.

All those errors can be measured over different subsets of the data. In this regard, the **training error** refers to the error measured on the training samples, the **selection error** is measured on the selection samples, and the **testing error** is measured on the testing samples.

Activation Functions

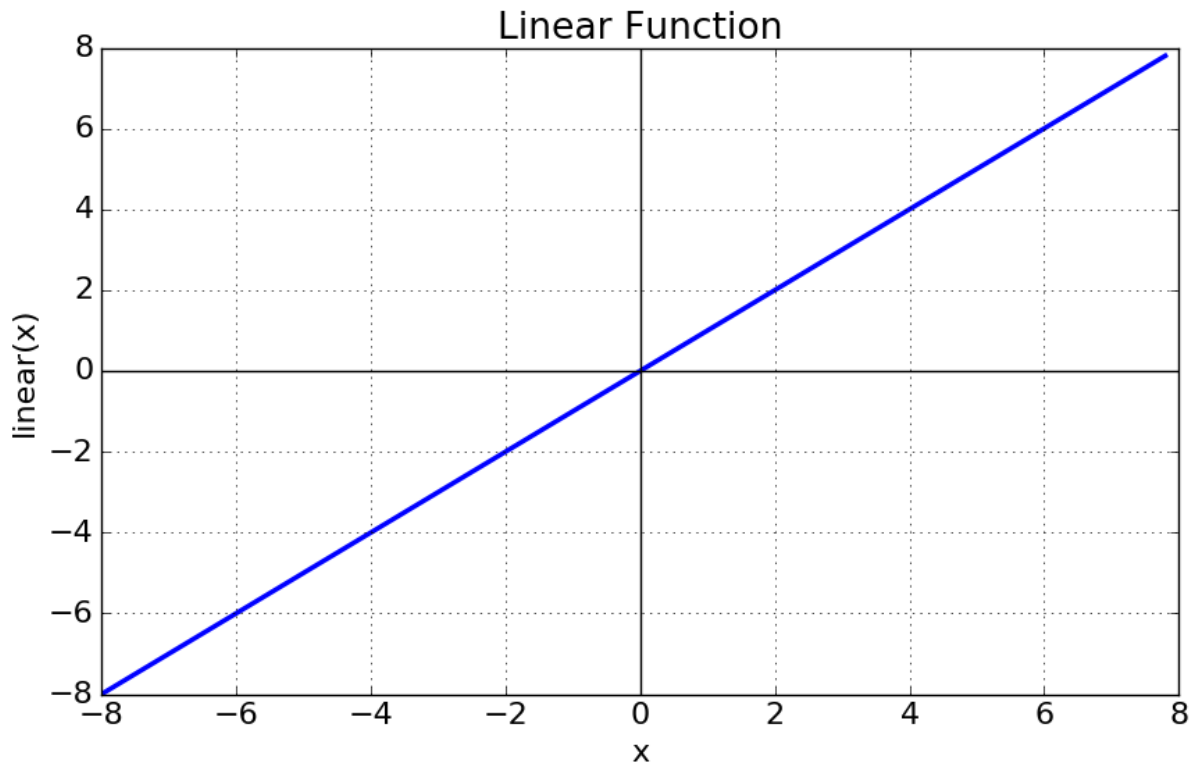
It is used to determine the output of neural network like yes or no. It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).

The Activation Functions can be basically divided into 2 types-

1. Linear Activation Function
2. Non-linear Activation Functions

Linear or Identity Activation Function

As you can see the function is a line or linear. Therefore, the output of the functions will not be confined between any range.



Equation : f(x) = x

Range : (-infinity to infinity)

It doesn't help with the complexity or various parameters of usual data that is fed to the neural networks.

Non-linear Activation Function

The Nonlinear Activation Functions are the most used activation functions. Nonlinearity helps to makes the graph look something like this

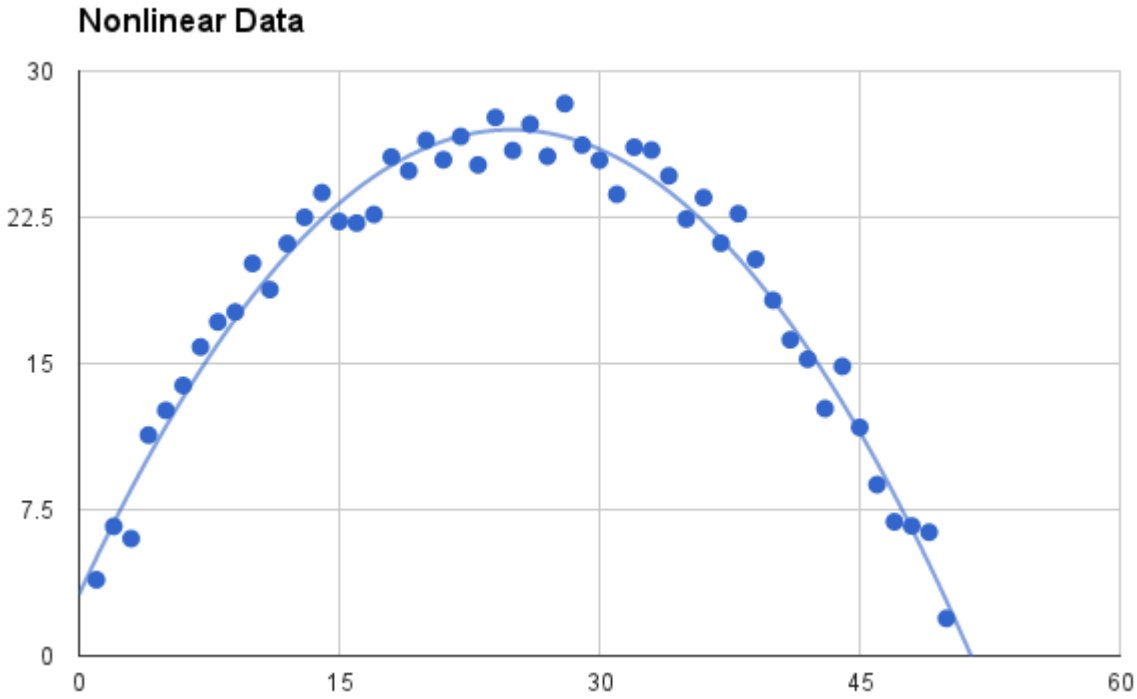


Fig: Non-linear Activation Function

It makes it easy for the model to generalize or adapt with variety of data and to differentiate between the output.

Gradient Descent for Machine Learning

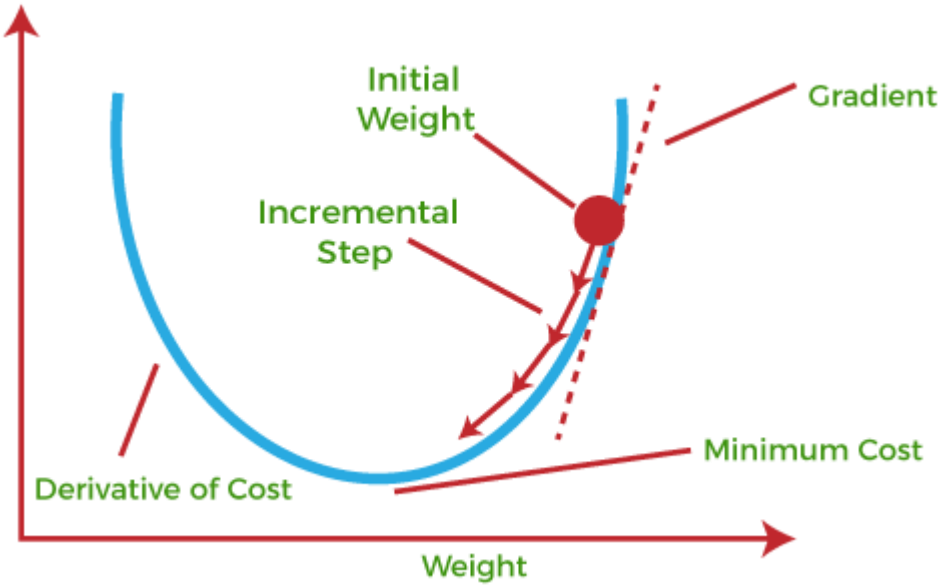
Gradient Descent is known as one of the most commonly used optimization algorithms to train machine learning models by means of minimizing errors between actual and expected results. Further, gradient descent is also used to train Neural Networks.

In machine learning, optimization is the task of minimizing the cost function parameterized by the model's parameters. The main objective of gradient descent is to minimize the convex function using iteration of parameter updates. Once these machine learning models are optimized, these models can be used as powerful tools for Artificial Intelligence and various computer science applications.

It helps in finding the local minimum of a function.

The best way to define the local minimum or local maximum of a function using gradient descent is as follows:

- If we move towards a negative gradient or away from the gradient of the function at the current point, it will give the **local minimum** of that function.
- Whenever we move towards a positive gradient or towards the gradient of the function at the current point, we will get the **local maximum** of that function.



This entire procedure is known as Gradient Ascent, which is also known as steepest descent. ***The main objective of using a gradient descent algorithm is to minimize the cost function using iteration.*** To achieve this goal, it performs two steps iteratively:

- Calculates the first-order derivative of the function to compute the gradient or slope of that function.
- Move away from the direction of the gradient, which means slope increased from the current point by alpha times, where Alpha is defined as Learning Rate. It is a tuning parameter in the optimization process which helps to decide the length of the steps.

What is Cost-function?

The cost function is defined as the measurement of difference or error between actual values and expected values at the current position and present in the form of a single real number. It helps to increase and improve machine learning efficiency by providing feedback to this model so that it can minimize error and find the local or global minimum. Further, it continuously iterates along the direction of the **negative gradient** until the cost function approaches zero. At this steepest descent point, the model will stop learning further. Although cost function and loss function are considered synonymous, also there is a minor difference between them. The slight difference between the loss function and the cost function is about the error within the training of machine learning models, as **loss function refers to the error of one training example, while a cost function calculates the average error across an entire training set.**

The cost function is calculated after making a hypothesis with initial parameters and modifying these parameters using gradient descent algorithms over known data to reduce the cost function.

Types of Gradient Descent

Based on the error in various training models, the Gradient Descent learning algorithm can be divided into **Batch gradient descent, stochastic gradient descent, and mini-batch gradient descent.** Let's understand these different types of gradient descent:

1. Batch Gradient Descent:

Batch gradient descent (BGD) is used to **find the error for each point i**n the training set and update the model after evaluating all training examples. This procedure is known as the training epoch. In simple words, it is a greedy approach where we have to sum over all examples for each update.

Advantages of Batch gradient descent:

- It produces less noise in comparison to other gradient descent.
- It produces stable gradient descent convergence.
- It is Computationally efficient as all resources are used for all training samples.

2. Stochastic gradient descent

Stochastic gradient descent (SGD) is a type of gradient descent **that runs one training example per iteration**. Or in other words, it processes a training epoch for each example within a dataset and updates each training example's parameters one at a time. **As it requires only one training example at a time, hence it is easier to store in allocated memory.** However, it shows some computational efficiency losses in comparison to batch gradient systems as it shows frequent updates that require more detail and speed. Further, due to frequent updates, it is also treated as a noisy gradient. However, sometimes it can be helpful in finding the global minimum and also escaping the local minimum.

Advantages of Stochastic gradient descent:

In Stochastic gradient descent (SGD), learning happens on every example, and it consists of a few advantages over other gradient descent.

- It is easier to allocate in desired memory.
- It is relatively fast to compute than batch gradient descent.
- It is more efficient for large datasets.

3. MiniBatch Gradient Descent:

Mini Batch gradient descent is the combination of both batch gradient descent and stochastic gradient descent. It divides the training datasets into small batch sizes then performs the updates on those batches separately. Splitting training datasets into smaller batches make a balance to maintain the computational efficiency of batch gradient descent and speed of stochastic gradient descent. Hence, we can achieve a special type of gradient descent with higher computational efficiency and less noisy gradient descent.

Advantages of Mini Batch gradient descent:

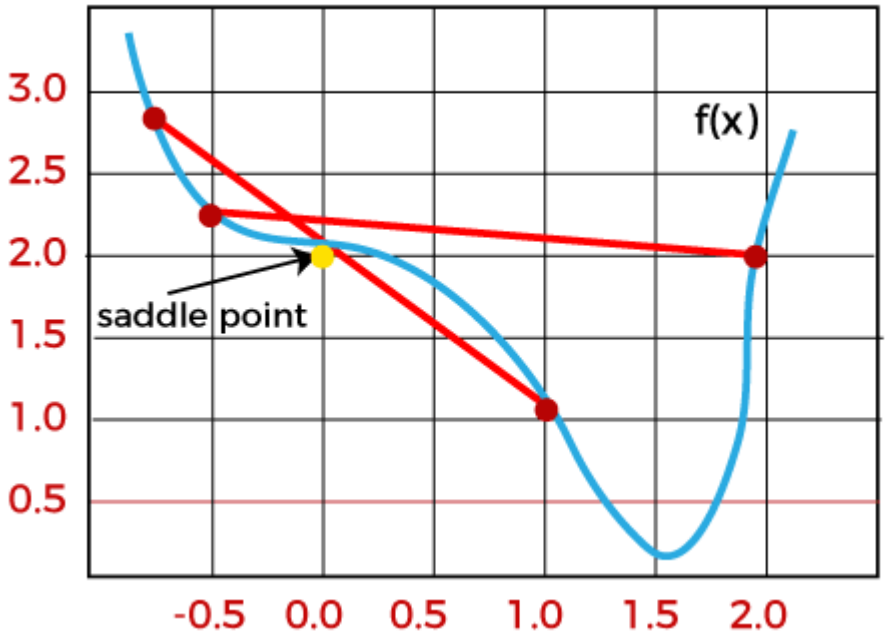
- It is easier to fit in allocated memory.
- It is computationally efficient.
- It produces stable gradient descent convergence.

Challenges with the Gradient Descent

Although we know Gradient Descent is one of the most popular methods for optimization problems, it still also has some challenges. There are a few challenges as follows:

1. Local Minima and Saddle Point:

For convex problems, gradient descent can find the global minimum easily, while for non-convex problems, it is sometimes difficult to find the global minimum, where the machine learning models achieve the best results.



Whenever the slope of the cost function is at zero or just close to zero, this model stops learning further. Apart from the global minimum, there occur some scenarios that can show this slop, which is saddle point and local minimum. Local minima generate the shape similar

to the global minimum, where the slope of the cost function increases on both sides of the current points.

In contrast, with saddle points, the negative gradient only occurs on one side of the point, which reaches a local maximum on one side and a local minimum on the other side. The name of a saddle point is taken by that of a horse's saddle.

The name of local minima is because the value of the loss function is minimum at that point in a local region. In contrast, the name of the global minima is given so because the value of the loss function is minimum there, globally across the entire domain the loss function.

2. Vanishing and Exploding Gradient

In a deep neural network, if the model is trained with gradient descent and backpropagation, there can occur two more issues other than local minima and saddle point.

Vanishing Gradients:

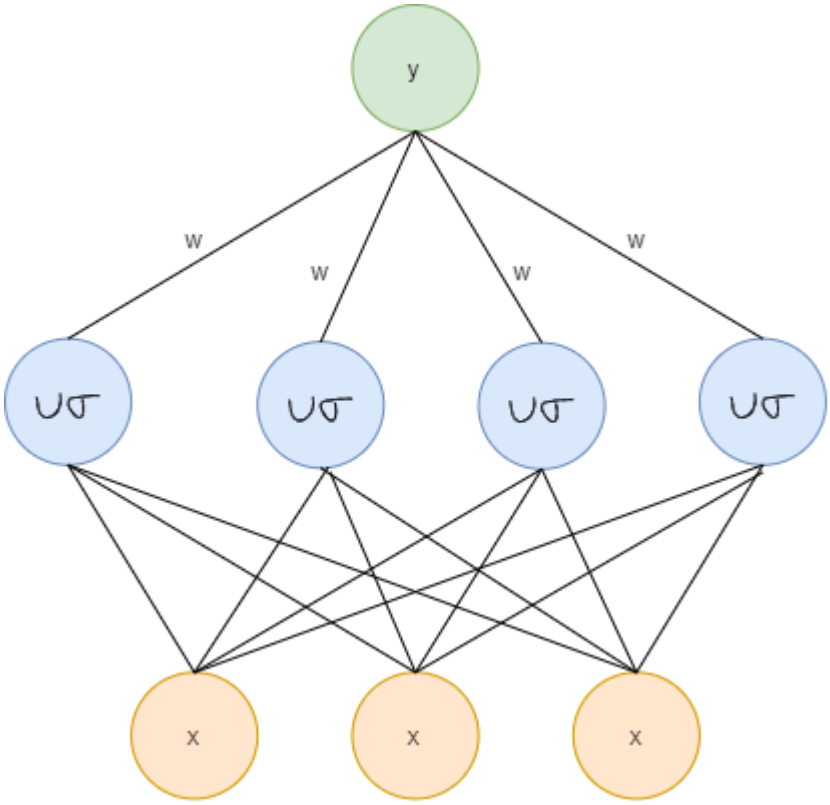
Vanishing Gradient occurs when the gradient is smaller than expected. During backpropagation, this gradient becomes smaller that causing the decrease in the learning rate of earlier layers than the later layer of the network. Once this happens, the weight parameters update until they become insignificant.

R**adial basis function**

R**adial basis function (RBF) networks** have a fundamentally different architecture than most neural network architectures. Most neural network architecture consists of many layers and introduces nonlinearity by repetitively applying nonlinear activation functions. RBF network on the other hand only consists of an input layer, a single hidden layer, and an output layer.

The input layer is not a computation layer, it just receives the input data and feeds it into the special hidden layer of the RBF network. The computation that is happened inside the hidden layer is very different from most neural networks, and this is where the power of the RBF

network comes from. The output layer performs the prediction task such as classification or regression.



Hopfield network

The Hopfield Neural Networks, invented by Dr John J. Hopfield consists of one layer of *'n'* fully connected recurrent neurons. It is generally used in performing auto association and optimization tasks. It is calculated using a converging interactive process and it generates a different response than our normal neural nets.

Discrete Hopfield Network: It is a fully interconnected neural network where each unit is connected to every other unit. It behaves in a discrete manner, i.e. it gives finite distinct output, generally of two types:

- **Binary (0/1)**
- **Bipolar (-1/1)**

The weights associated with this network is symmetric in nature and has the following properties.

1. $w_{ij} = w_{ji}$
2. $w_{ii} = 0$

Structure & Architecture

- Each neuron has an inverting and a non-inverting output.
- Being fully connected, the output of each neuron is an input to all other neurons but not self.

Fig 1 shows a sample representation of a Discrete Hopfield Neural Network architecture having the following elements.

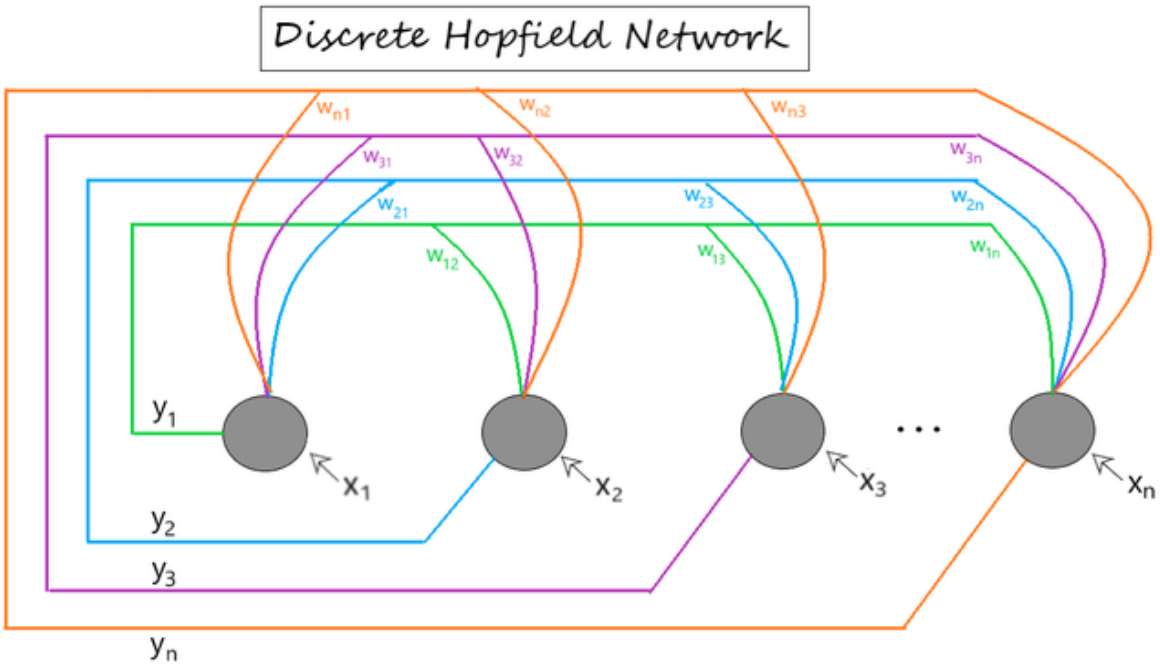


Fig 1: Discrete Hopfield Network Architecture

[x1 , x2 , … , xn] -> Input to the n given neurons.
[y1 , y2 , … , yn] -> Output obtained from the n given neurons
W_{ij} -> weight associated with the connection between the i^{th} and the j^{th} neuron.

Training Algorithm

For storing a set of input patterns S(p) [p = 1 to P], where **$S(p) = S_1(p) \dots S_i(p) \dots S_n(p)$**, the weight matrix is given by:

- **For binary patterns**

$$w_{ij} = \sum_{p=1}^{P}[2s_i(p) - 1][2s_j(p) - 1] \ (w_{ij} \ for \ all \ i \neq j)$$

- **For bipolar patterns**

$$w_{ij} = \sum_{p=1}^{P}[s_i(p)s_j(p)] \ (where \ w_{ij} = 0 \ for \ all \ i = j)$$

(i.e. weights here have no self-connection)

Continuous Hopfield Network: Unlike the discrete hopfield networks, here the time parameter is treated as a continuous variable. So, instead of getting binary/bipolar outputs, we can obtain values that lie between 0 and 1. It can be used to solve constrained optimization and associative memory problems. The output is defined as:

$v_i = g(u_i)$

where,

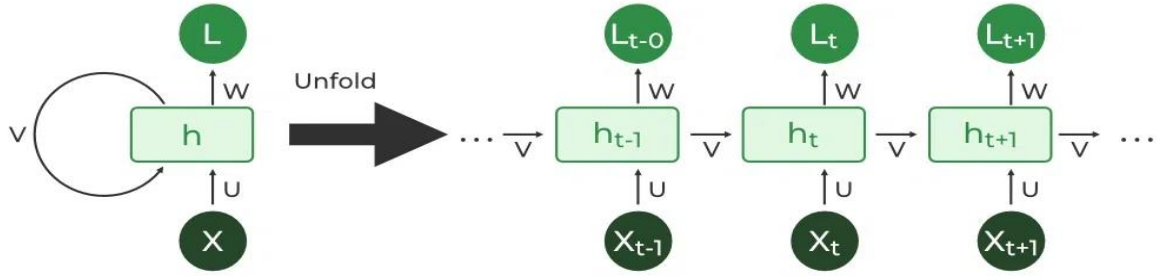
v_i = output from the continuous hopfield network
u_i = internal activity of a node in continuous hopfield network.

The name given to identifying feasible solutions out of a very large set of candidates, where the problem can be modeled in terms of arbitrary constraints.

Recurrent Neural Network(RNN)

Recurrent Neural Network(RNN) is a type of Neural Network where the output from the previous step is fed as input to the current step.

The main and most important feature of RNN is its **Hidden state**, which remembers some information about a sequence. The state is also referred to as *Memory State* since it remembers the previous input to the network. It uses the same parameters for each input as it performs the same task on all the inputs or hidden layers to produce the output. This reduces the complexity of parameters, unlike other neural networks.



How RNN works

The Recurrent Neural Network consists of multiple fixed activation function units, one for each time step. Each unit has an internal state which is called the hidden state of the unit. This hidden state signifies the past knowledge that the network currently holds at a given time step. This hidden state is updated at every time step to signify the change in the knowledge of the network about the past. The hidden state is updated using the following recurrence relation:-

The formula for calculating the current state:

$$h_t = f(h_{t-1}, x_t)$$

where:

h_t -> current state
h_{t-1} -> previous state
x_t -> input state

Formula for applying Activation function(tanh):

$$h_t = \tanh (W_{hh}h_{t-1} + W_{xh}x_t)$$

where:

w_{hh} -> weight at recurrent neuron
w_{xh} -> weight at input neuron

The formula for calculating output:

$$y_t = W_{hy}h_t$$

Y_t -> output
W_{hy} -> weight at output layer

Advantages of Recurrent Neural Network

1. An RNN remembers each and every piece of information through time. It is useful in time series prediction only because of the feature to remember previous inputs as well. This is called Long Short Term Memory.
2. Recurrent neural networks are even used with convolutional layers to extend the effective pixel neighborhood.

Disadvantages of Recurrent Neural Network

1. Gradient vanishing and exploding problems.
2. Training an RNN is a very difficult task.
3. It cannot process very long sequences if using tanh or relu as an activation function.

Applications of Recurrent Neural Network

1. Language Modelling and Generating Text
2. Speech Recognition
3. Machine Translation
4. Image Recognition, Face detection
5. Time series Forecasting

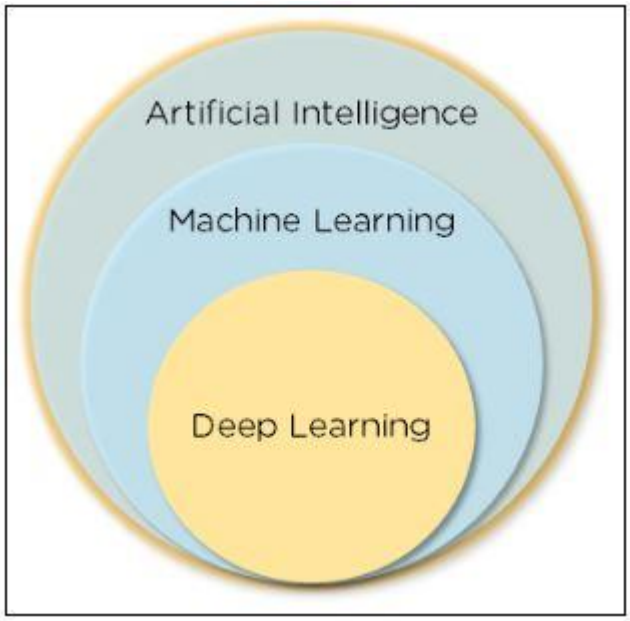
Types Of RNN

There are four types of RNNs based on the number of inputs and outputs in the network.

1. One to One
2. One to Many
3. Many to One
4. Many to Many

DEEP LEARNING

Artificial intelligence is the ability of a machine to imitate intelligent human behavior. Machine learning allows a system to learn and improve from experience automatically. Deep learning is an application of machine learning that uses complex algorithms and deep neural nets to train a model.

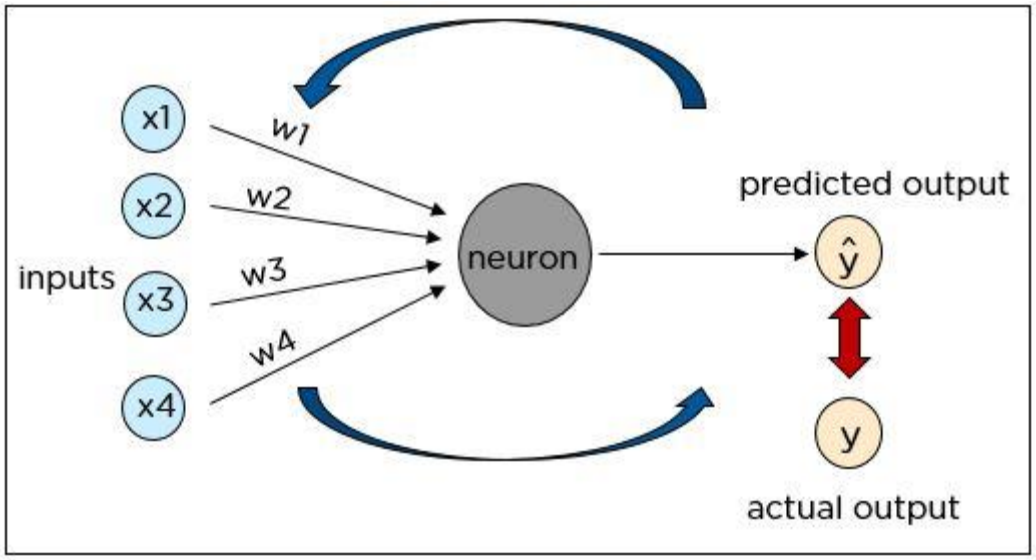


Importance of Deep Learning

- Machine learning works only with sets of structured and semi-structured data, while deep learning works with both structured and unstructured data
- Deep learning algorithms can perform complex operations efficiently, while machine learning algorithms cannot
- Machine learning algorithms use labeled sample data to extract patterns, while deep learning accepts large volumes of data as input and analyzes the input data to extract features out of an object
- The performance of machine learning algorithms decreases as the number of data increases; so to maintain the performance of the model, we need a deep learning

Cost Function

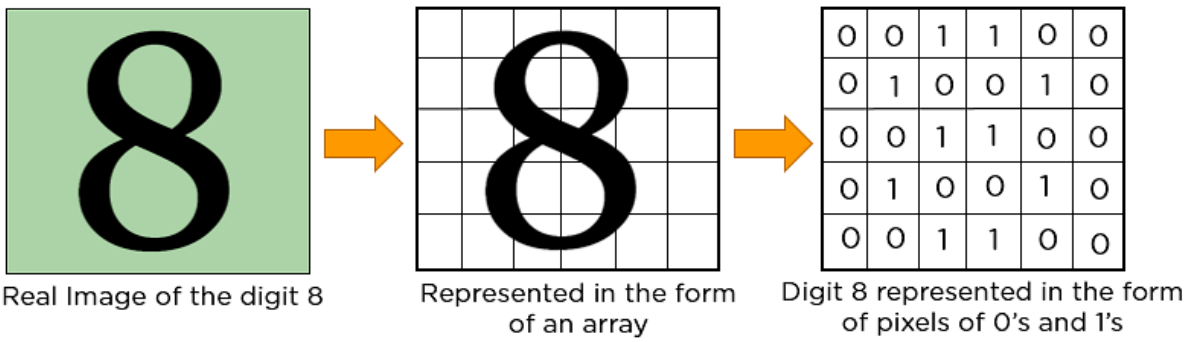
The cost function is one of the significant components of a neural network. The cost value is the difference between the neural nets predicted out;pput and the actual output from a set of labeled training data. The least-cost value is obtained by making adjustments to the weights and biases iteratively throughout the training process.



CONVOLUTIONAL NEURAL NETWORK

A convolutional neural network is a feed-forward neural network that is generally used to analyze visual images by processing data with grid-like topology. It's also known as a ConvNet. A convolutional neural network is used to detect and classify objects in an image.

In CNN, every image is represented in the form of an array of pixel values.



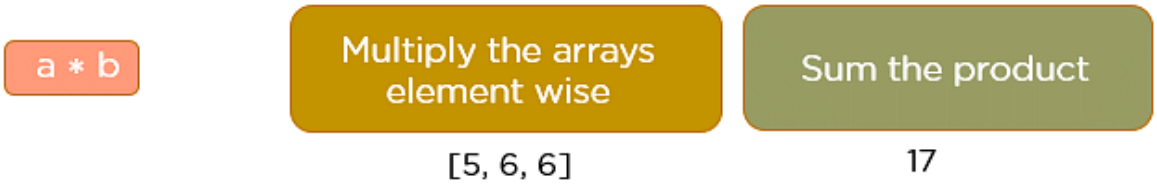
The convolution operation forms the basis of any convolutional neural network. Let's understand the convolution operation using two matrices, a and b, of 1 dimension.

a = [5,3,2,5,9,7]

b = [1,2,3]

In convolution operation, the arrays are multiplied element-wise, and the product is summed to create a new array, which represents a*b.

The first three elements of the matrix a are multiplied with the elements of matrix b. The product is summed to get the result.

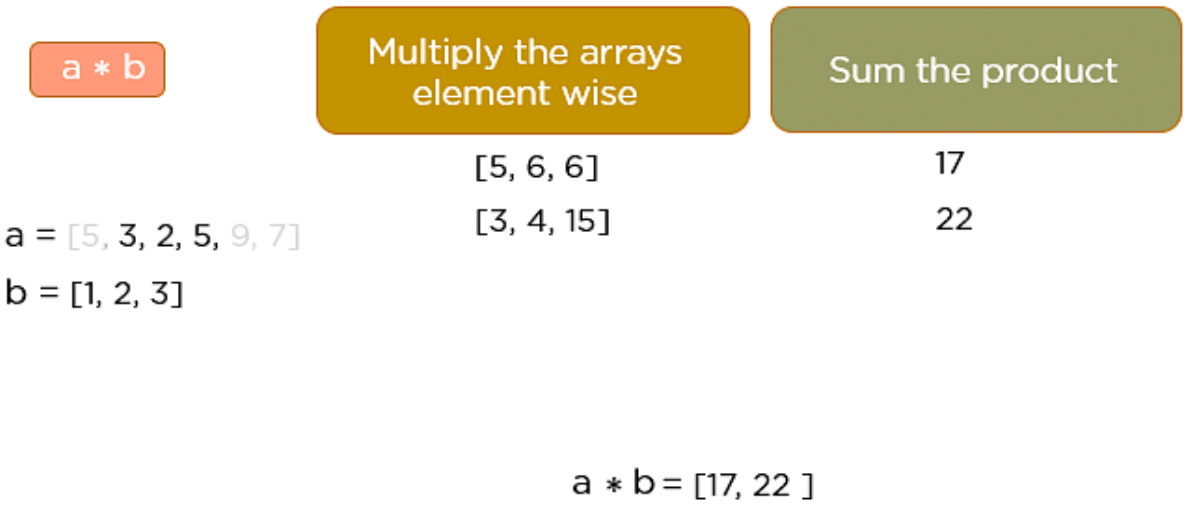


a = [5, 3, 2, 5, 9, 7]

b = [1, 2, 3]

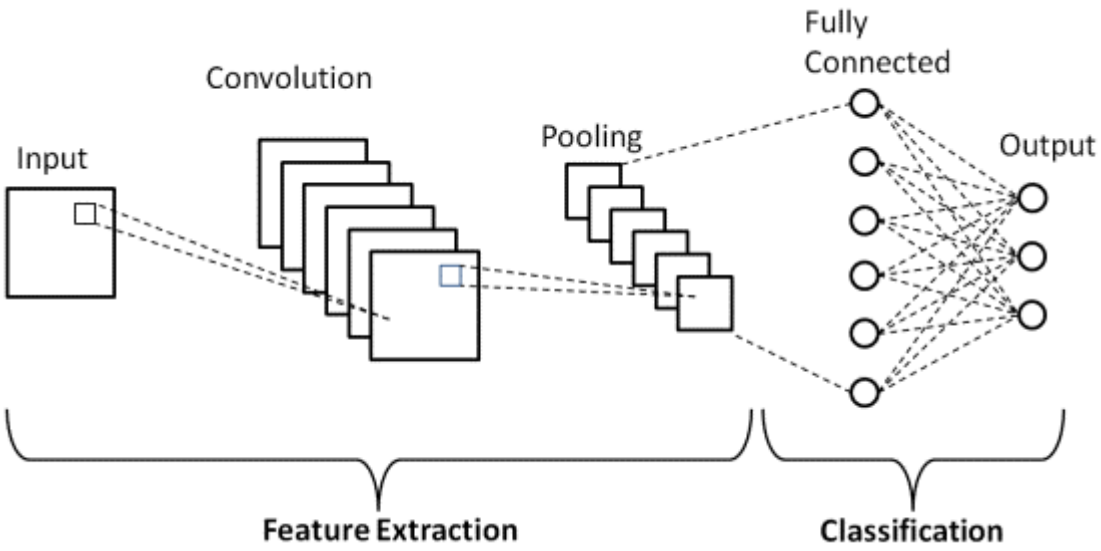
a * b = [17,]

The next three elements from the matrix a are multiplied by the elements in matrix b, and the product is summed up.



This process continues until the convolution operation is complete.

CNN Architecture



Layers in a Convolutional Neural Network

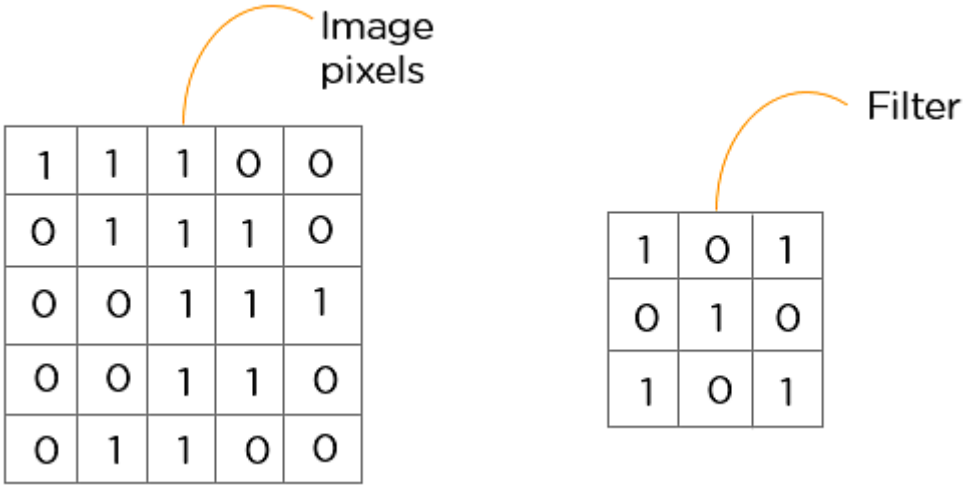
A convolution neural network has multiple hidden layers that help in extracting information from an image. The four important layers in CNN are:

1. Convolution layer
2. ReLU layer
3. Pooling layer
4. Fully connected layer

Convolution Layer

This is the first step in the process of extracting valuable features from an image. A convolution layer has several filters that perform the convolution operation. Every image is considered as a matrix of pixel values.

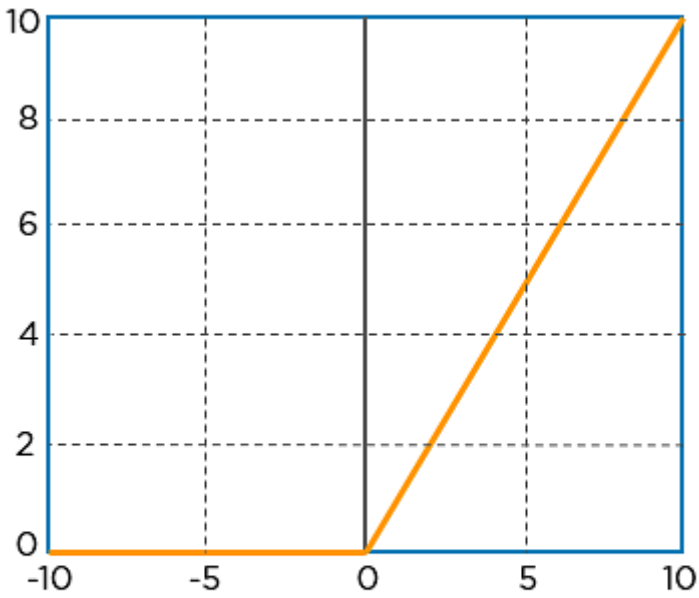
Consider the following 5x5 image whose pixel values are either 0 or 1. There's also a filter matrix with a dimension of 3x3. Slide the filter matrix over the image and compute the dot product to get the convolved feature matrix.



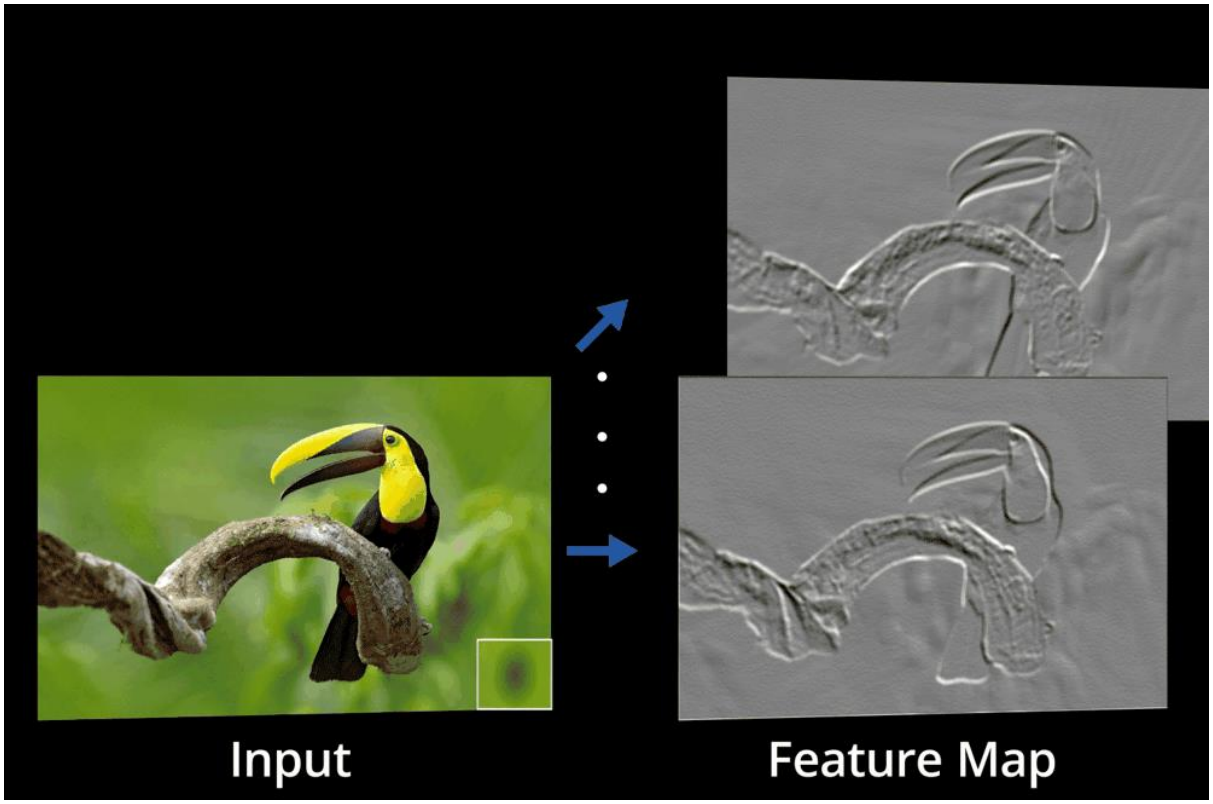
ReLU layer

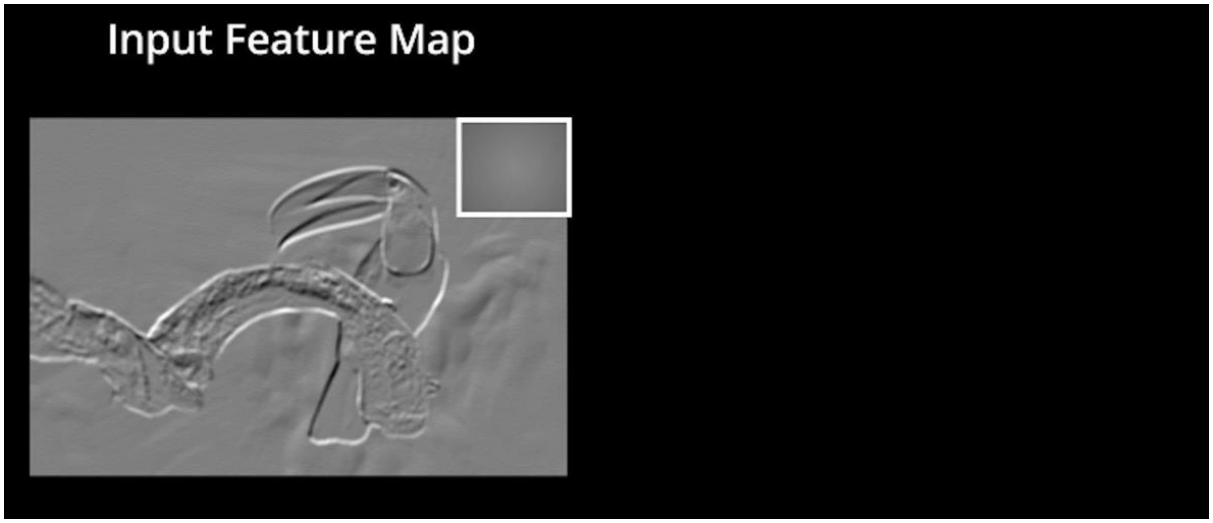
ReLU stands for the rectified linear unit. Once the feature maps are extracted, the next step is to move them to a ReLU layer.

ReLU performs an element-wise operation and sets all the negative pixels to 0. It introduces non-linearity to the network, and the generated output is a rectified feature map. Below is the graph of a ReLU function:



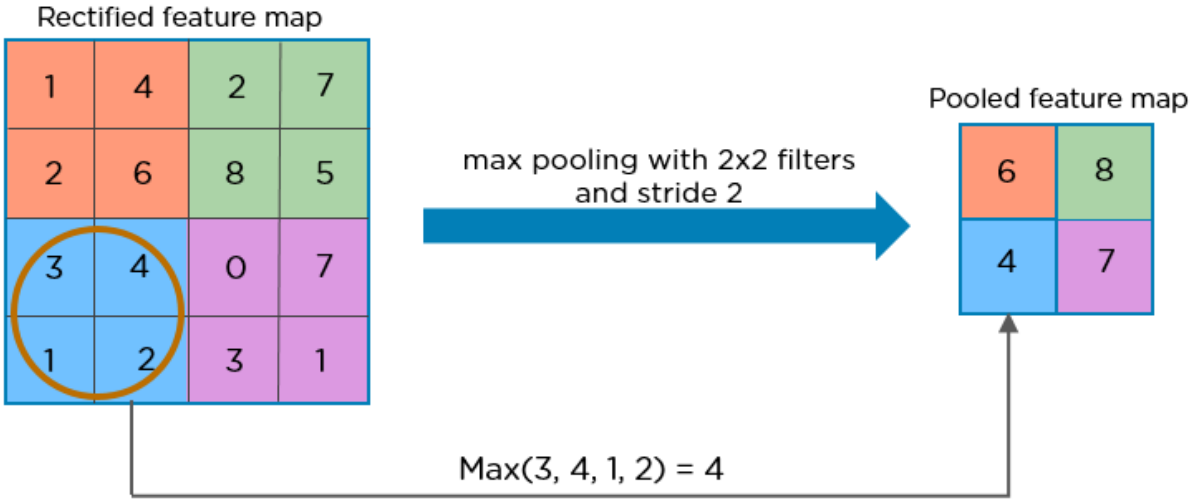
The original image is scanned with multiple convolutions and ReLU layers for locating the features.



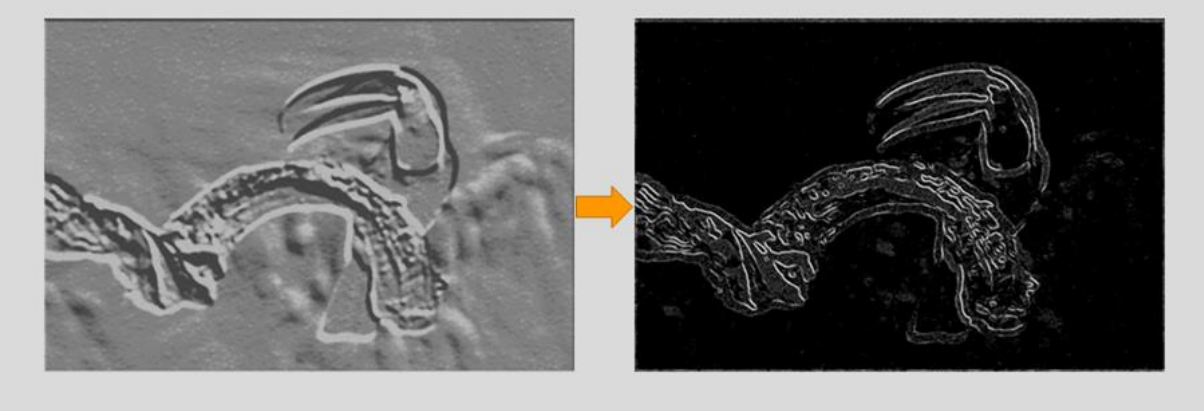


Pooling Layer

Pooling is a down-sampling operation that reduces the dimensionality of the feature map. The rectified feature map now goes through a pooling layer to generate a pooled feature map.



The pooling layer uses various filters to identify different parts of the image like edges, corners, body, feathers, eyes, and beak.

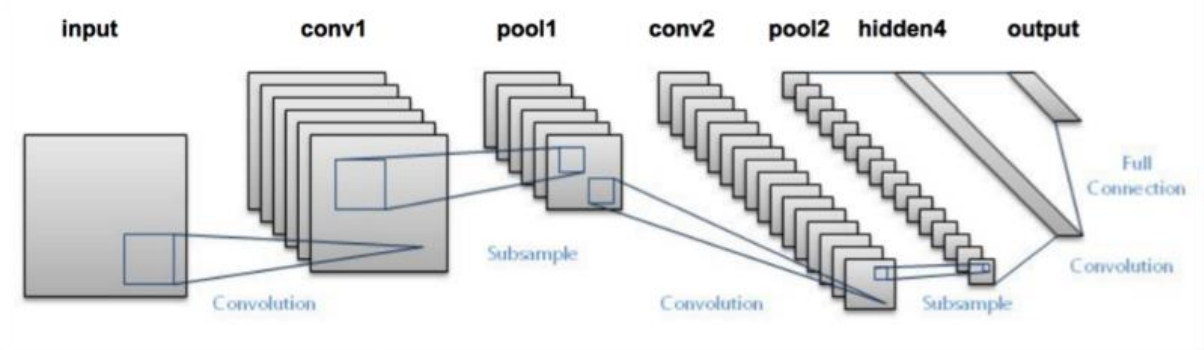


Fully Connected Layer

The Fully Connected (FC) layer consists of the weights and biases along with the neurons and is used to connect the neurons between two different layers. These layers are usually placed before the output layer and form the last few layers of a CNN Architecture.

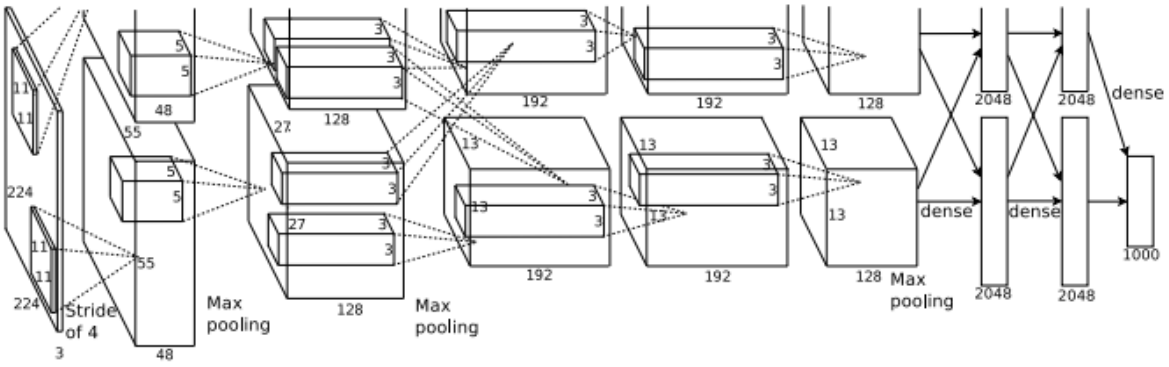
LeNet-5 (1998)

LeNet-5, a pioneering 7-level convolutional network by LeCun et al in 1998, that classifies digits, was applied by several banks to recognise hand-written numbers on checks (cheques) digitized in 32x32 pixel greyscale input images. The ability to process higher resolution images requires larger and more convolutional layers, so this technique is constrained by the availability of computing resources.



AlexNet (2012)

In 2012, AlexNet significantly outperformed all the prior competitors and won the challenge by reducing the top-5 error from 26% to 15.3%. The second place top-5 error rate, which was not a CNN variation, was around 26.2%.



The network had a very similar architecture as LeNet by Yann LeCun et al but was deeper, with more filters per layer, and with stacked convolutional layers. It consisted 11x11, 5x5,3x3, convolutions, max pooling, dropout, data augmentation, ReLU activations, SGD with momentum. It attached ReLU activations after every convolutional and fully-connected layer. AlexNet was trained for 6 days simultaneously on two Nvidia Geforce GTX 580 GPUs which is the reason for why their network is split into two pipelines. AlexNet was designed by the SuperVision group, consisting of Alex Krizhevsky, Geoffrey Hinton, and Ilya Sutskever.
